

MediTrust

Sprint_2_Backlog

Group7

The goal of **Sprint 2** is to extend the MediTrust system by adding a machine learning model for cardiovascular risk prediction and integrating it with the existing frontend and backend system.

In Sprint 1 we built the basic system with login, signup, database storage, and a simple rule-based risk calculation. In this sprint, we focused on replacing that dummy logic with a trained machine learning model and connecting it to the frontend interface.

The system should now allow users to enter clinical parameters and receive a predicted cardiovascular risk level generated by the ML model.

User Stories

1. Machine Learning Prediction

As a clinician, I want the system to predict cardiovascular risk using a machine learning model so that the assessment becomes more realistic.

2. Model Training

As a developer, I want to train machine learning models using the dataset so that the system can generate predictions based on real patterns.

3. Prediction API

As a system, I want an API endpoint that takes patient clinical data and returns the predicted risk level.

4. Risk Assessment Interface

As a clinician, I want a form where I can enter patient clinical parameters to evaluate the risk level.

5. GitHub Branch Workflow

As a development team, we want to use feature branches in GitHub so that both members can work on different parts of the system without conflicts.

Task ID	Task Description	Assigned To	Priority	Est. Time	Status
T1	Create feature branches for Sprint 2 development	Ravi	High	1 hr	Completed
T2	Study and prepare cardiovascular dataset	Ravi	High	4 hrs	Completed
T3	Implement data preprocessing script	Ravi	High	3 hrs	Completed
T4	Train machine learning models	Ravi	High	5 hrs	Completed
T5	Evaluate models and select best one	Ravi	High	2 hrs	Completed
T6	Implement risk stratification logic	Ravi	High	2 hrs	Completed
T7	Create FastAPI prediction endpoint (/predict)	Ravi	High	3 hrs	Completed
T8	Improve frontend risk assessment page	Uday	High	5 hrs	Completed
T9	Connect frontend form to prediction API	Uday	High	3 hrs	Completed
T10	Test prediction workflow end-to-end	Uday	High	2 hrs	Completed
T11	Merge feature branches into sprint branch	Ravi	Medium	1 hr	Completed
T12	Prepare Sprint 2 documentation	Ravi	Medium	3 hrs	Completed

All planned tasks for Sprint 2 were completed successfully.

At the end of this sprint, the system now supports:

- Machine learning based cardiovascular risk prediction
- A frontend interface to enter clinical parameters
- FastAPI prediction endpoint
- Integration between frontend and ML model
- GitHub branch based development workflow

Project management GitHub Board link: <https://github.com/users/RavindraSSK/projects/7>